Applied Probability and Statistics

Dhirubhai Ambani University

Course Syllabus

Course Title

Applied Probability and Statistics

Credit Structure

Lecture hours per week: 3
Tutorial hours per week: 0
Practical hours per week: 0
Total Credits: 3

Program/Semester

MTech ICT 1st Sem, BTech ICT 7th Sem

Prerequisites Courses

Calculus

Instructor

Dr. Abhishek Tilva

Course Description

This course provides a rigorous introduction to probability and its applications in stochastic modelling, statistics and data science. Many applied topics such as martingales, stochastic processes, regression, etc will be discussed.

Course Objectives and Outcomes

This course is designed to equip students with a rigorous understanding of probability theory and its applications in modeling uncertainty across real-world phenomena. The course aims to build students' ability to formulate, analyze, and implement probabilistic and statistical models relevant to domains such as data science, finance, and engineering.

Upon completion of this course, students will be able to:

- Develop understanding of concepts in probability and statistics and related notions such as stochastic processes.
- Utilize probabilistic reasoning and modeling tools in research and industry applications across fields such as finance, data science, operations research, and engineering.
- Apply statistical techniques such as regression and statistical design to model and interpret real-world data.

Course Materials/References

Suggested Textbooks

- Grimmett, G. R., and D. R. Sterzaker. ""Probability and Random Processes" 4th edition."
- DeGroot, Morris H., and Mark J. Schervish. ""Probability and statistics" 4th edition."
- Ramsey, Fred and Schafer, Daniel. "The Statistical Sleuth: A course in methods of Data Analysis" 3rd edition.
- F. E. Harrell, Jr. ""Regression Modelling Strategies with Applications to Linear Models, Logistic Regression and Survival analysis.""

Evaluation/Grading Policy

- Midterm: 25%
- Final: 75%

Detailed Course Contents**

** The contents are subject to change based on time and other factors.

Topic Name	**Content (2-3 lines per 4-6 lectures)**	**No. of Lectures**
Foundations of probability	Probability spaces, σ-fields, conditional probability, independence, random variables and vectors, Borel-Cantelli Lemmas	3
Discrete and continuous random variables	Cdf, Pmf, Pdf, moments, common discrete and continuous distributions, conditional distributions and expectation, functions of random variables	4
Generating functions and applications	Different types of generating functions, branching processes, convergence and asymptotics, large deviations	6
Martingales	Martingales, sub and supermartingales, optional stopping, random walk, wald's identity	6
Stochastic processes	Brownian motion – definition and properties, Donsker's theorem, Reflection principle, Black Scholes Model, Poisson process – definition and properties, superposition, thinning, poisson process in higher dimensions	6
Statistical inference	Method of moments, Maximum likelihood estimation, Bayesian inference, Sampling distributions, Confidence intervals, non parametric density estimation	8
Regression and machine learning	Supervised and unsupervised learning algorithms, kNN regression and classification, k-means clustering, linear regression	4